

DEMYSTIFYING ANDROID APP SECURITY

SCHEDULE

Date	Theory Session 09:30–11:00	Lab 11:30–01:00	Theory Session 02:00–03:30	Lab 04:00–05:30
Day 1 – 21 Sep	i) Latest Mobile Application Security Threats and Vulnerabilities ii) Introduction to Android Apps • Introduction To Mobile Apps • The App Sandbox and the Permission Model • Android Security Architecture • Android File System	Setting Up the Android Development Environment	Android Application Development – Fundamentals • Application lifecycle • Activities, Services, Broadcast Receivers & Content Providers • Intents and IPC • Android application structure • Modern Android development with Kotlin & Jetpack Compose	Android Development Environment & Application Building • Android Studio setup • SDK/Emulator configuration • Create a Kotlin/Jetpack Compose application • Build, install and debug APK • Explore APK structure
Day 2 – 22 Sep	Android Application Analysis – Static Analysis • APK anatomy • DEX, resources and manifest • Decompilation & reverse engineering concepts • Code obfuscation • Static security analysis methodology	Static Analysis of Android Applications • JADX • APKTool • MobSF • Android Studio APK Analyzer • Manifest analysis • Code & resource analysis	Android Application Analysis – Dynamic Analysis • Dynamic analysis methodology • ADB fundamentals • Runtime behaviour analysis • Network traffic analysis • SSL/TLS and certificate validation • SSL Pinning concepts • Runtime instrumentation	Dynamic Analysis Lab • ADB-based analysis • Logcat analysis • Traffic interception • Runtime application observation • Instrumentation-based analysis • Identify and document security weaknesses

	- Identifying hardcoded secrets - Insecure storage & exported components - Weak cryptography - Misconfigurations	Identify security findings and prepare observations	Root/emulator/debug detection concepts	
Day 3 – 23 Sep	i) OWASP Top 10 Mobile threats ii) Secure Coding Practices - Verify for Security Early and Often - Parameterized Queries - Encode Data - Validate All Inputs	Survey on Vulnerabilities Detected in Various Popular Mobile Apps	Secure Coding Practices - Implement Identity and Authentication Controls - Implement Access Controls - Cryptographically Secure Data - Implement Logging and Intrusion Detection - Leverage Security Frameworks and Libraries - Monitor Error and Exception Handling	Sample Code Snippets for App Vulnerabilities
Day 4 – 24 Sep	iOS Application Development – Fundamentals - iOS ecosystem & architecture - Swift fundamentals - Application lifecycle - View/UI architecture - iOS application sandbox - Entitlements & permissions	iOS Development Basics - Xcode environment - Create basic Swift application - Build and run application - Explore application bundle - Basic debugging and signing concepts	iOS Application Security & Analysis - iOS application security architecture - IPA structure - Static analysis methodology - Dynamic analysis methodology - Keychain analysis - Binary analysis	iOS Application Analysis Lab - IPA/application bundle analysis - Static analysis tools - Runtime observation - Network analysis - Identify common iOS security weaknesses

	• Keychain • Secure storage • App Transport Security • iOS application security model		• Network security • Jailbreak detection concepts • Runtime instrumentation concepts	
Day 5 – 25 Sep	Advanced Mobile Security – Case Studies • Mobile attack chains • Application tampering & repackaging • Runtime manipulation • Reverse engineering challenges • Root/jailbreak & instrumentation bypass concepts • Secure API design • Device trust & attestation • Mobile threat modelling	Advanced Security Analysis – Case Studies • End-to-end analysis of a sample application • Identify attack surface • Static + dynamic analysis • Demonstrate vulnerability • Develop mitigation recommendations	Challenges, Research Directions & Emerging Mobile Security • Limitations of mobile security controls • Android vs iOS security models • Platform-level security research • Mobile malware & emerging threats • AI-assisted mobile security analysis • Future directions in mobile security • Research opportunities in AOSP and mobile platforms	Capstone Exercise / Assessment • End-to-end security assessment • Analyse a sample application • Identify vulnerabilities • Prepare security assessment report • Group discussion & presentation • Feedback and concluding session